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Network delay inference from additive metrics

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Published Random Structures and Algorithms (2010)

DOI https://doi.org/10.1002/rsa.20305

Paper Link https://arxiv.org/abs/math/0604367

Domain Signal processing, statistics & applied mathematics

Abstract

We use computational phylogenetic techniques to solve a central problem in inferential network monitoring. More precisely, we design a novel algorithm for multicast-based delay inference, that is, the problem of reconstructing delay characteristics of a network from end-to-end delay measurements on network paths. Our inference algorithm is based on additive metric techniques used in phylogenetics. It runs in polynomial time and requires a sample of size only poly(log n). We also show how to recover the topology of the routing tree.

Real-World Impact & Application

Cloud Data Centers & VoIP Quality Monitoring: In hyperscale cloud environments (like AWS and Microsoft Azure), transient latency jitter degrades VoIP calls and database queries. Tsang et al. demonstrated that network delays can be modeled as additive tree metrics and solved with linear algebra, allowing network monitoring tools to locate slow fiber links without deploying probes inside the core switching fabric.